

Ricky Kumar

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[GitHub](#) · [LinkedIn](#) · [LeetCode](#) · [Portfolio](#)

Summary

M.Tech Artificial Intelligence student focused on building agentic and RAG-based LLM systems that automate decision-making and reduce hallucination. Experienced in evaluating LLM outputs with LangSmith, optimizing fine-tuning (PEFT, QLoRA, transfer learning) to cut training cost, and orchestrating multi-agent, multilingual (text, voice, video) pipelines for enterprises handling sensitive data and policy documents.

Education

M.Tech in Artificial Intelligence, IIIT Pune

Aug 2025 – May 2027

CGPA: 7.81 · Coursework: Gen-AI, Agentic AI, Deep L, Machine L, Big Data Analytics, Crypto currency and Blockchain, RM

B.Tech in Computer Science, MAIT Solan

Aug 2018 – May 2022

CGPA: 7.7 · Coursework: Data structure and Algo, C/C++, Python, Web dev, ML/DL, OOPs, COA, CN, CD, OS, TOC, Linux, DM, EM, SE, Distributed Comp, Soft Computing, Indian Constitution

Projects

1. Multi-Agent Hotel Booking Assistant – Taj Aguada (AWS Bedrock)

Jun – Jul 2026

- Architected a multi-agent booking system on Amazon Bedrock Agents, orchestrating sub-agents for room search, availability, booking confirmation, and guest FAQs.
- Built a Bedrock Knowledge Base (Titan Embeddings, ~300-token chunks, 15% overlap) over policies, tariffs, and amenities for accurate retrieval.
- Used Claude on Bedrock as the orchestrator to parse intent, trigger Lambda action groups for live availability/booking, and generate responses; persisted session and booking state in DynamoDB.
- Benchmarked Claude/Titan token pricing and tuned chunk size and top-*k* to optimize answer quality per API cost.
- Applied Bedrock Guardrails for PII masking, denied-topic filtering, and knowledge-grounded responses to eliminate hallucinated booking details.

2. FinBot – Conversational Loan Advisor

Jan – Apr 2026

- Built a RAG chatbot on a FAISS vector store covering loan policies, terms, regulations, and regional law, with document validation..
- Designed a 3-layer hallucination defense: source-grounded citations, confidence-based web search fallback (Tavily), and output validation to block impossible claims.
- Stored per-user conversation history in PostgreSQL to enable context-aware follow-ups and guardrail tuning.
- Achieved 95%+ factual accuracy across 1,000+ test queries with zero hallucinations on ground-truth data; added prompt-injection prevention and PII masking.

3. UPSC Study Assistant – Corrective RAG with Local LLM/LLM API

Mar – May 2026

- Ingested study material (PDFs via PyPDF2, lecture audio via Whisper, figures via CLIP, and diagrams via OCR) into a FAISS vector store for unified multimodal retrieval.
- Built a corrective RAG pipeline in which a local open-source Llama model scores each retrieved chunk for relevance, discards low-quality results, and reranks the remaining documents before generation.
- Used Redis for short-term session memory and PostgreSQL to track long-term learning patterns and user habits, enabling personalized study recommendations.

4. Phishing Detection Model

Aug – Dec 2025

- Built a layer that intercepts suspicious any kind of web link, email links and executes them in an isolated Selenium sandbox before they reach the user's browser.
- Detected fake login forms, hidden password fields, and malicious API calls via real-time DOM and JavaScript behavioral analysis, blocking credential harvesting.
- Performed various data pipeline like preprocessing, cleaning, modeling, deploying.

Technical Skills

Languages: C, C++, Python, SQL

Databases & Retrieval: MySQL, MongoDB, DynamoDB, FAISS, PostgreSQL, Redis

ML/DL Tools: PyTorch, NumPy, Pandas, scikit-learn, Matplotlib, Jupyter, Google Colab

DL Architectures: CNN, RNN, ANN, Encoder-Decoder, U-Net

Models: GAN, VAE, Random Forest, Logistic/Linear Regression, KNN, DBSCAN, Decision Tree, SVM

Generative AI & LLM: LangGraph, LangChain, CrewAI, LangSmith, Self-Corrective RAG, Graph-RAG, Fine-

tuning (RLHF, QLoRA, PEFT), Ollama, Whisper, CLIP, Prompt Engineering, LLM Guardrails

Backend & Deployment: FastAPI, MCP, AWS SageMaker

Safety & Monitoring: MLOps, Bias Detection, Compliance Checks, Prompt Injection Prevention

Certifications

- Agentic AI on AWS (Bedrock Agents, CrewAI, MCP) – Udemy
- Complete AI, Machine Learning & Data Science – Udemy

Experience & Achievements

Teaching Assistant, IIIT Pune – Lab evaluation and mentoring for OOPs, DSA, DBMS, Cloud Computing, and Operating Systems

GATE 2025: All India Rank ~10,000 out of 170,000 in Computer Science (Score: 448/1000)

LeetCode: 200+ problems solved across data structures and algorithms